Multivariate Calculus Lecture Notes
Section 12.2 – Vectors in Three Dimensions

I. Right-handed coordinate system
   a. The standard 3D xyz-coordinate system looks like this:

   b. It obeys what is called the right-hand rule: If you point the fingers of your right hand in the direction of the first variable and curl them in the direction of the second variable then your thumb points in the direction of the third variable.
   c. Some 3D vector calculus formulas depend on having a right-handed system.

II. Distance formula
   a. How do you measure the distance between two points (x, y) and (a, b) in the plane?
      i. Graphically:

      ii. Essentially, this involves the Pythagorean Theorem: $d = \sqrt{(x-a)^2 + (y-b)^2}$.
   b. How do you measure the distance between two points (x, y, z) and (a, b, c) in space?
      i. Graphically:

      ii. This involves the Pythagorean Theorem twice.
         1. We have $(AB)^2 = (AC)^2 + (BC)^2$.
         2. And then we have $(AC)^2 = (AD)^2 + (CD)^2$.
         3. Putting it together we have $d = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$.
   c. Equation of a sphere
      i. The distance formula can be used to derive the equation of a sphere.
      ii. A sphere is defined as the set of all points equidistant from a given point.
         1. The given point we will call (a, b, c).
         2. The fixed distance we will call r.
      iii. Derivation
         1. So, if a point (x, y, z) is on the sphere, it must satisfy the expression $r = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$, or $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$, since all quantities are positive.
         2. And only points that satisfy this equation can be on the sphere.
         3. Thus the equation for a sphere with center (a, b, c) and radius r is $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$.
      iv. Comment

1. This is only the surface of the sphere.
2. To include the interior we would have $(x-a)^2+(y-b)^2+(z-c)^2 \leq r^2$.

v. Example: Find the center and radius of the following sphere:

$$2x^2+2y^2+2z^2+4x+8y-4z=6.$$

1. Divide by 2: $x^2+y^2+z^2+2x+4y-2z=3$.
2. Next, complete the square:

$$x^2+2x+1+y^2+4y+4+z^2-2x+1=3+1+4+1$$
$$\implies (x+1)^2+(y+2)^2+(z-1)^2=9.$$

3. Thus the center is $(-1,-2,1)$, and the radius is 3.

d. Equation of a cylinder
   i. The graph of $(x-a)^2+(y-b)^2=r^2$ in 2D is a circle of radius r centered at (a,b).
   ii. What is the same graph in 3D?
      1. Since this equation holds true regardless of the value of z, we can extend the circle arbitrarily up or down in the z direction while still getting points that satisfy it.
      2. So we're getting a cylinder whose central axis is a vertical line through $(a,b,0)$.

III. 3D vector basics
   a. Many of the 2D vector properties extend in a straightforward way to 3D vector properties.
   b. A 3D vector $\mathbf{v}$ has *components* given by $\mathbf{v}=\langle a,b,c\rangle$.
   c. The *length* or *magnitude* of a 3D vector is given by the distance formula: $||\mathbf{v}|| = \sqrt{a^2+b^2+c^2}$.
   d. Two 3D vectors $\mathbf{u}$ and $\mathbf{v}$ are *parallel* if $\mathbf{u}=c\mathbf{v}$ for some scalar c.
   e. The *standard basis vectors* in 3D are $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.
      i. In component form, these are $\mathbf{i}=\langle 1,0,0\rangle$, $\mathbf{j}=\langle 0,1,0\rangle$, $\mathbf{k}=\langle 0,0,1\rangle$.
      ii. A generic 3D vector $\mathbf{v}=\langle a,b,c\rangle$ can be expressed in terms of the standard basis vectors as $\mathbf{v}=a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$.
      iii. You'll talk more about this kind of thing in linear algebra (such as switching to other basis vectors).

IV. Lines in three dimensions
   a. Basic ideas
      i. How do you represent a line in the plane?
         1. Two points
         2. Point and slope
      ii. Slope in space
         1. In the plane, slope is used to represent lines.
         2. In space, though, vectors are a more convenient way to represent slope.
         3. In fact, what does slope mean in space?
            • There are both a z/x slope and a z/y slope, for instance.

iii. In space, it's convenient to represent a line with a point and a direction.
   1. Direction is supplied by a vector $\mathbf{v}$.
   2. For example:

iv. The line can then be expressed as the original point P plus all multiples of $\mathbf{v}$.

b. Line equations
   i. Vector form
      1. Let $\mathbf{r}_0$ be the vector whose endpoint is the point $P_0 = (x_0, y_0, z_0)$.
      2. Let $\mathbf{v}$ be a direction vector, $\mathbf{v} = \langle a, b, c \rangle$.
      3. Then the line through the point P in the direction of $\mathbf{v}$ can be expressed, in vector form, as $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$, $-\infty < t < \infty$, i.e., the position vector $\mathbf{r}_0$ plus all scalar multiples of the vector $\mathbf{v}$.
      4. Here, $\mathbf{r}(t)$ is the position vector of some point (x, y, z) on the line.
      5. Alternatively, this may be expressed as $\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t\langle v_1, v_2, v_3 \rangle$, $-\infty < t < \infty$.
   ii. Parametric form
      1. $x = x_0 + tv_1$
      2. $y = y_0 + tv_2$
      3. $z = z_0 + tv_3$
   iii. What would be the difference between the lines $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$ and $\mathbf{r}(t) = \mathbf{r}_0 + t(2\mathbf{v})$?
      1. They're the same line, graphically-speaking.
      2. However, if we think of $\mathbf{r}(t)$ as the position of an object at time t, an object traveling according to the second equation is traveling twice as fast as an object traveling according to the first one.

c. Examples
   i. Find parametric equations for the line through the point $(5, 1, 3)$ and parallel to the vector $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.
      - $x = 5 + t$, $y = 1 + 2t$, $z = 3 - t$.
   ii. Find parametric equations for the line through $(2, 4, -3)$ and $(3, -1, 1)$.
      1. A vector in the direction of the line is $\mathbf{v} = \langle 1, -5, 4 \rangle$.
      2. So equations are given by $x = 2 + t$, $y = 4 - 5t$, $z = -3 + 4t$.

d. How do you tell if two lines are parallel?
   i. The direction vectors are multiples of each other.
   ii. Example: Is the line through $(1, 2, 3)$ and $(4, 7, 1)$ parallel to the line through $(5, 1, 3)$ and $(-1, -9, 7)$?
      1. Direction vector for first line: $\langle 3, 5, -2 \rangle$
      2. Direction vector for second line: $\langle -6, -10, 4 \rangle$
      3. So the lines are parallel.

e. How do you tell if two lines intersect?
   i. If there is a solution to the line equations, they intersect, and if not, they don't.
   ii. Example: Do the lines $x = t$, $y = -2 + 3t$, $z = 4 - t$ and $x = 2s$, $y = 3 + s$, $z = -3 + 4s$ intersect?
      1. See if there is a solution to the equations $t = 2s$, $-2 + 3t = 3 + s$, $4 - t = -3 + 4s$.

- First equation into second equation: $-2 + 3(2s) = 3 + s \Rightarrow s = 1$.
- First equation into third equation: $4 - (2s) = -3 + 4s \Rightarrow s = 7/6$.

2. We get two different values for s, and thus the lines do not intersect.

V. Midpoint formula

a. What's the point halfway between $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$?

i. Geometrically:

ii. We have $\overrightarrow{OM} = \overrightarrow{OP_1} + \frac{1}{2}\overrightarrow{P_1P_2} = \overrightarrow{OP_1} + \frac{1}{2}(\vec{OP_2} - \overrightarrow{OP_1}) = \frac{1}{2}(\overrightarrow{OP_1} + \overrightarrow{OP_2})$.

iii. Thus the midpoint is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right).$$

b. For example, the midpoint of the points $(1, 0, 2)$ and $(-3, 6, 1)$ is $(-1, 3, 1\frac{1}{2})$.

Homework: Section 12.2 #6, 9, 10, 13, 20, 31, 33, 38, 40, 42, 48, 52, 53, 61.